

Vocabulary

Parent function: the simplest form of a family — no shifts, stretches, or reflections.

Domain / Range: domain is the set of allowed x -values; range is the set of resulting y -values.

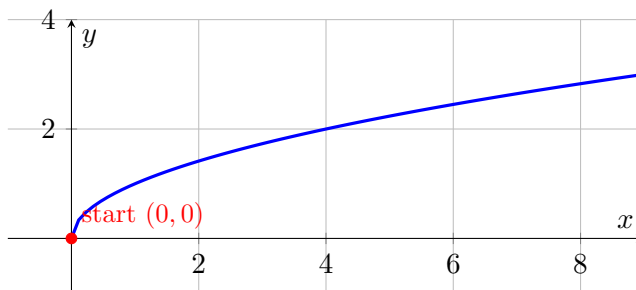
We do together. Name the parent of $f(x) = \sqrt{x}$ and give its domain.

Step 1. Match the rule to a known family: this is the square root parent.

Step 2. Recall its shape: it starts at the point $(0, 0)$ and curves up to the right.

Step 3. A square root needs a non-negative inside, so the domain is $x \geq 0$.

See it:



$y = \sqrt{x}$ — starts at the origin, domain $x \geq 0$.

You Try. Solve each. Show every step, then name the answer.

a) Name the parent of $f(x) = \frac{1}{x}$ and give its domain.

reciprocal (rational); domain $x \neq 0$

b) Name the parent of $f(x) = |x|$ and give its range.

absolute value; range $y \geq 0$

Summary (fill in): To name a parent: match the rule to a family, recall its shape, then state the domain.